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Enterprise Fleet Telematics Lakehouse & Dynamic Logistics Warehouse

SIC 8 · 2026

Big Data Engineering Track

Project 03 · Supply Chain & Intelligent Logistics

Agenda

- **Team**
- Introduction
- Problem Statement
- Methodology
- Results and Insights
- Conclusion
- Q&A

Our Team



Shahesta Mohamed

Source generation and ingestion

- Python fleet generator
- FastAPI ingestion API
- PostgreSQL, Debezium CDC and Kafka
- Bridge to Eventstream, hourly extractor



Ahmed Mohamed

Silver processing and anomaly detection

- Schema parsing and validation
- Deduplication and enrichment
- Movement analytics
- Anomaly rules and episodes



Yahya Moustafa

Gold model

- Star schema: dimensions and facts
- SCD Type 2 driver history
- Three KPI tables
- Quality checks and reconciliation



Abdullah Fathy

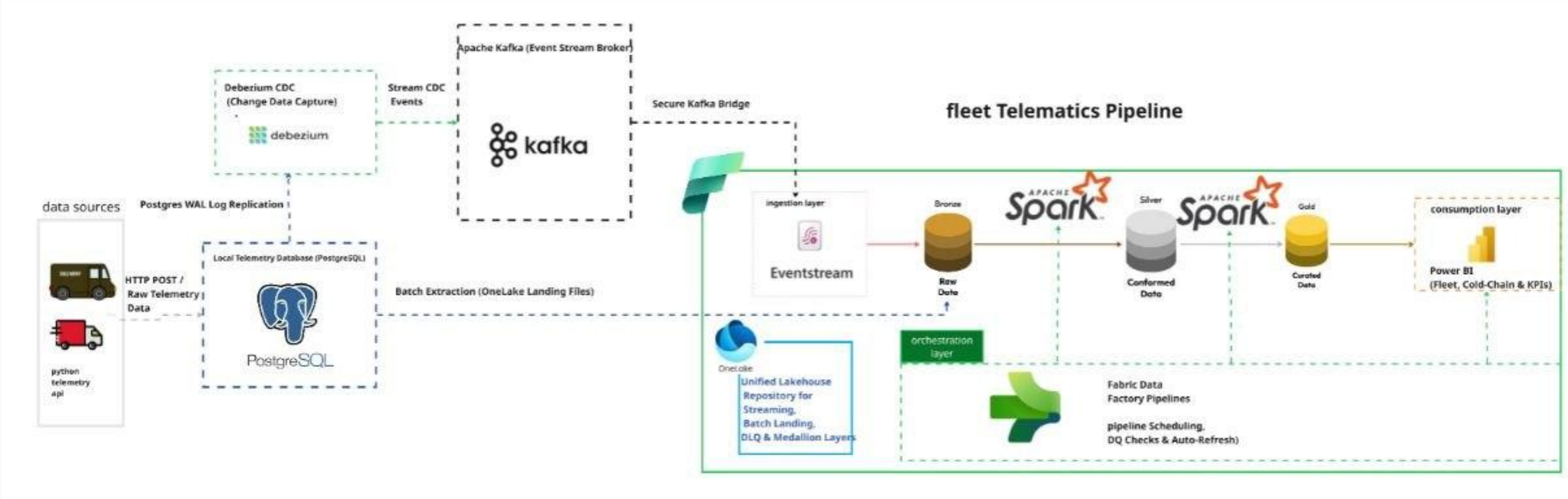
BI Development

- Six-page Power BI report
- Semantic model and DAX measures
- Orchestration and integration

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Project at a Glance: End-to-End Architecture



Hybrid design: local source systems in Docker → Microsoft Fabric lakehouse (Bronze · Silver · Gold) → Power BI

Introduction: A Data Platform for Fleet Visibility

5,000
vehicles
1,500 refrigerated · 3,500 dry vans

10
depots
across Egypt

30
fixed corridors
planned routes between depots

30 s
reading interval
GPS, engine, fuel, cargo temp

What we built

- A Medallion lakehouse on Microsoft Fabric
- A local source system: API → PostgreSQL → Kafka
- Streaming and batch ingestion into the cloud
- A Power BI report and a real-time alerts dashboard

Project objectives

- Repeatable data with known, injected problems
- Keep every record: nothing silently lost
- Clean Silver data; count each event once
- Trusted Gold KPIs: quality checks + reconciliation

Data notice: all fleet data is simulated. Results describe this simulation, not real fleet incidents.

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Problem Statement: Disconnected Signals

Signals arrive separately

GPS position and speed

Where is each vehicle, and how fast?

Engine and fuel (OBD-II)

Is the engine healthy and efficient?

Cargo temperature

Is refrigerated cargo still safe?

What managers need to know

1. Are vehicles using fuel sensibly?
2. Is cold cargo staying between 2–8 °C?
3. Which drivers idle too long or speed?
4. Where and when do unusual events happen?

Without a joined view, route deviations, refrigeration breaches and excessive idling are hard to investigate consistently.

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Data Sources: A Simulated Fleet with Injected Problems

Python generator (seed 42)

Fleet	5,000 vehicles · 5,200 drivers
Network	10 depots · 30 corridors · 10,000 planned trips
Readings	Every 30 s: GPS, speed, RPM, engine temp, fuel rate, odometer
Cargo	Cargo temperature on refrigerated trucks (2–8 °C)
Format	JSON Lines · UTC timestamps · metric units
Reproducible	Same seed and settings give the same output

Deliberate problems to catch

Cold-chain breach

cargo above 2–8 °C

Excessive idling

engine on, vehicle still

Route deviation

GPS off the corridor

Engine overheating

added by our team

Plus bad data records

- Malformed JSON
- Missing / null field
- Wrong type
- Impossible value
- Duplicate
- Late event

All data is simulated. Each injected problem is saved in a hidden truth file that detection code never reads.

Technology Stack and Why We Chose It

Python generator

Reproducible data with known problems

FastAPI + PostgreSQL

One entry point; insert-only source database

Debezium + Kafka

Streams new database rows without changing the API

Fabric Eventstream

Managed cloud entry point for the live stream

OneLake

One shared store for landing files and Delta tables

Delta Lakehouse

ACID writes, MERGE loads and time travel

PySpark + Spark SQL

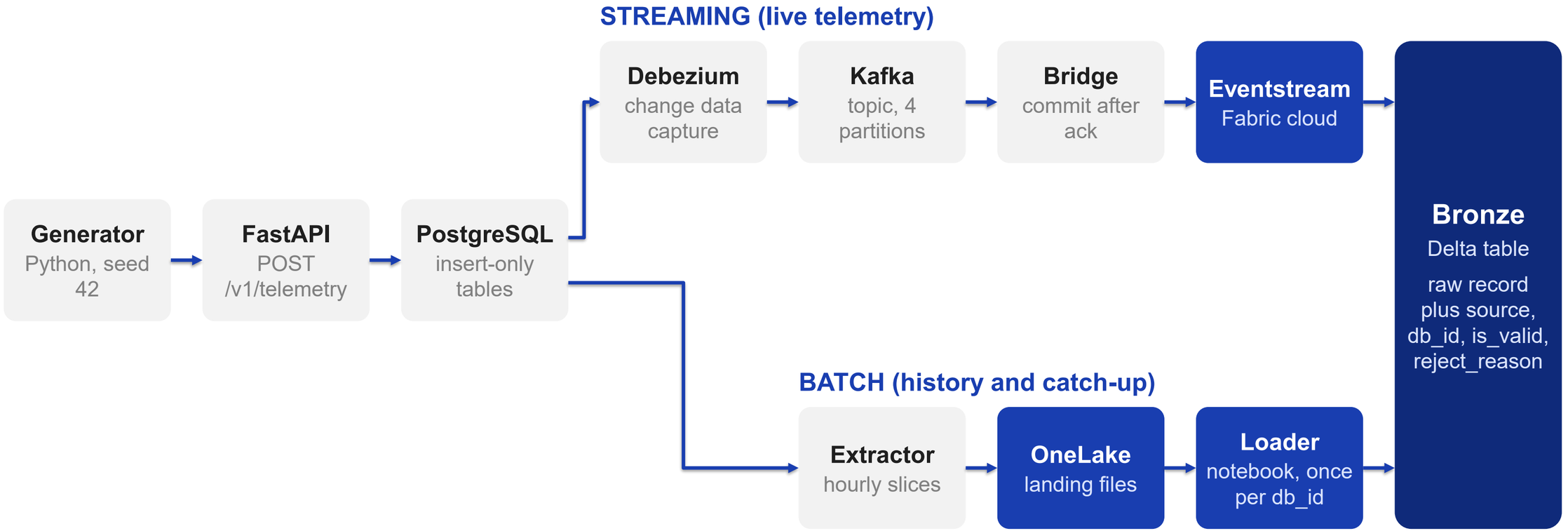
Distributed, explicit and auditable transforms

Power BI

Gold KPIs for managers, no SQL needed

Grey = local source system (Docker Compose) **Blue** = Microsoft Fabric cloud · Also in Fabric: pipelines, SQL analytics endpoint, real-time dashboard

Data Ingestion: Streaming and Batch Paths



Validated at the door

The API checks every line and stores it with is_valid and reject_reason.

Safe streaming

Offsets are committed only after the cloud acknowledges. Retries 1-2-4-8 s, then a dead-letter file.

Safe batch loads

One-hour slices tracked in extract_log. Re-running an hour never creates duplicates.

Medallion Lakehouse: Bronze → Silver → Gold

Bronze
Preserve raw evidence

- Every event stored as received
- Malformed records kept visible
- Metadata: source, db_id, run_id
- Insert-only

Silver
Validate, enrich, detect

- Parse with an explicit schema
- Rejects kept with a reason
- Deduplicate on db_id
- Enrich with vehicle data
- Detect anomaly episodes

Gold
Serve the business

- Star schema for Power BI
- Three KPI tables
- Driver history (SCD Type 2)
- Quality checks and reconciliation

Tables: bronze_telemetry_v2 → silver_telemetry, silver_rejected, silver_anomaly_episodes → gold_dim_*, gold_fact_*, gold_kpi_*

Engine: PySpark with Delta Lake. Gold uses idempotent MERGE writes, so re-running never creates duplicates. Power BI reads Gold only.

Silver Layer: Rules and Anomaly Episodes

Rule-based detection per reading

Rule	Condition
Idling	speed < 2 km/h and RPM > 0
Cold-chain violation	cargo temp < −2 °C or > 12 °C
Engine overheating	engine temp > 105 °C
Route-deviation signal	calculated speed > 100 km/h and jump > 0.5 km
RPM spike	RPM > 4,500

If several rules fire on one reading, the priority is: cold-chain, idling, route deviation, overheating, RPM spike. The Gold KPI uses the stricter 2–8 °C range.

From readings to episodes



≈ 40 flagged readings in a 20-minute idling stop (one every 30 s)



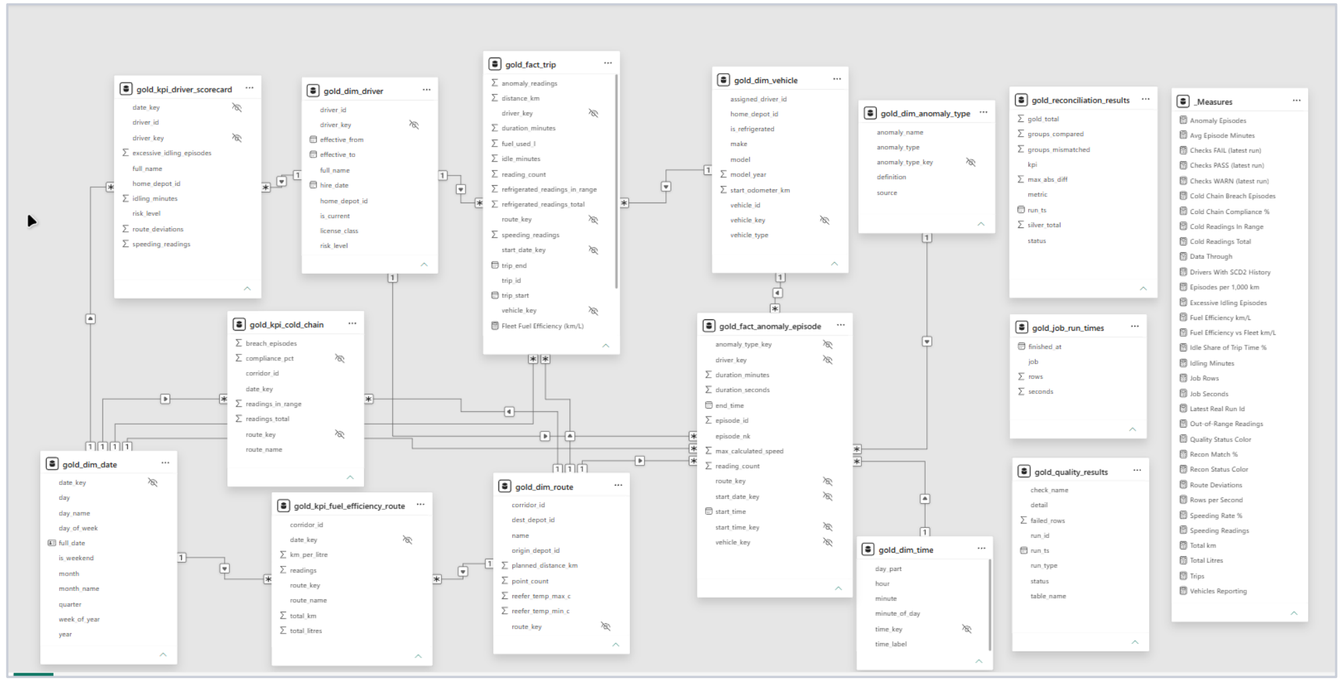
1 episode: start, end, duration, readings

A new episode starts after a gap of more than 15 minutes.

9,724 anomaly episodes in Silver

Check vs ground truth: precision 1.00, recall 0.83 (5 of 6 episodes) in one run. A second result exists; we will reconcile them.

Gold Layer: Star Schema and Business KPIs



Power BI semantic model built on the Gold tables

6 dimensions · 3 facts · 3 KPI tables	3 more tables log pipeline health
Driver history: SCD Type 2	416 drivers whose attributes changed
Ratios stored as numerator + denominator	Totals stay correct under any filter

Fuel efficiency (km/L)

total km ÷ total litres, by route and day

Cold-chain compliance (%)

readings at 2–8 °C ÷ all refrigerated readings

Driver scorecard

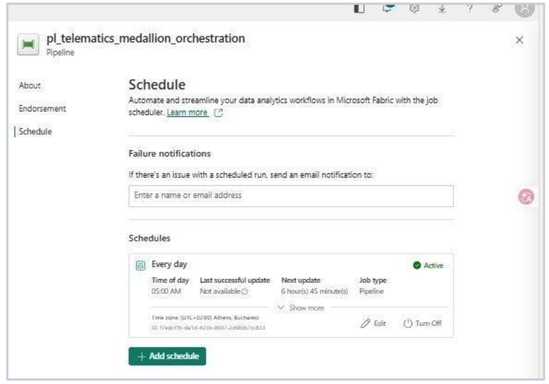
idling minutes, excessive-idling episodes, speeding readings (> 90 km/h), route deviations

Orchestration: Daily Run and Dashboard Refresh

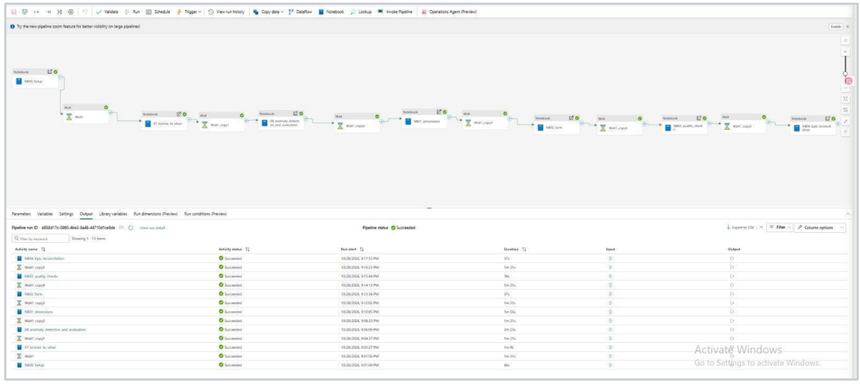
05:00 AM Cairo
Fabric pipeline: pl_telematics_medallion_orchestration

→

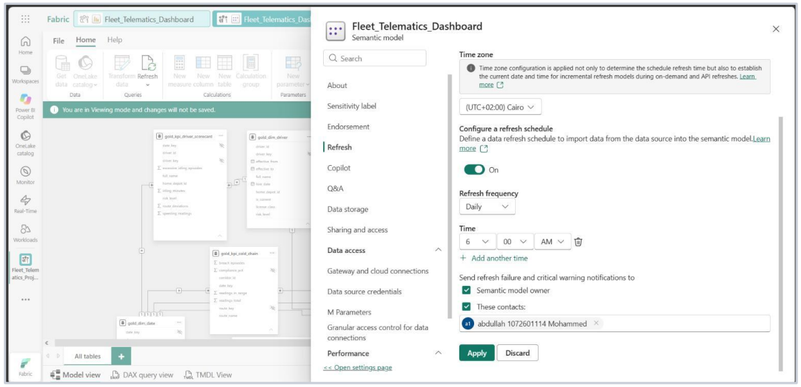
06:00 AM Cairo
Power BI semantic model refresh (Import mode)



Daily schedule



Successful pipeline run

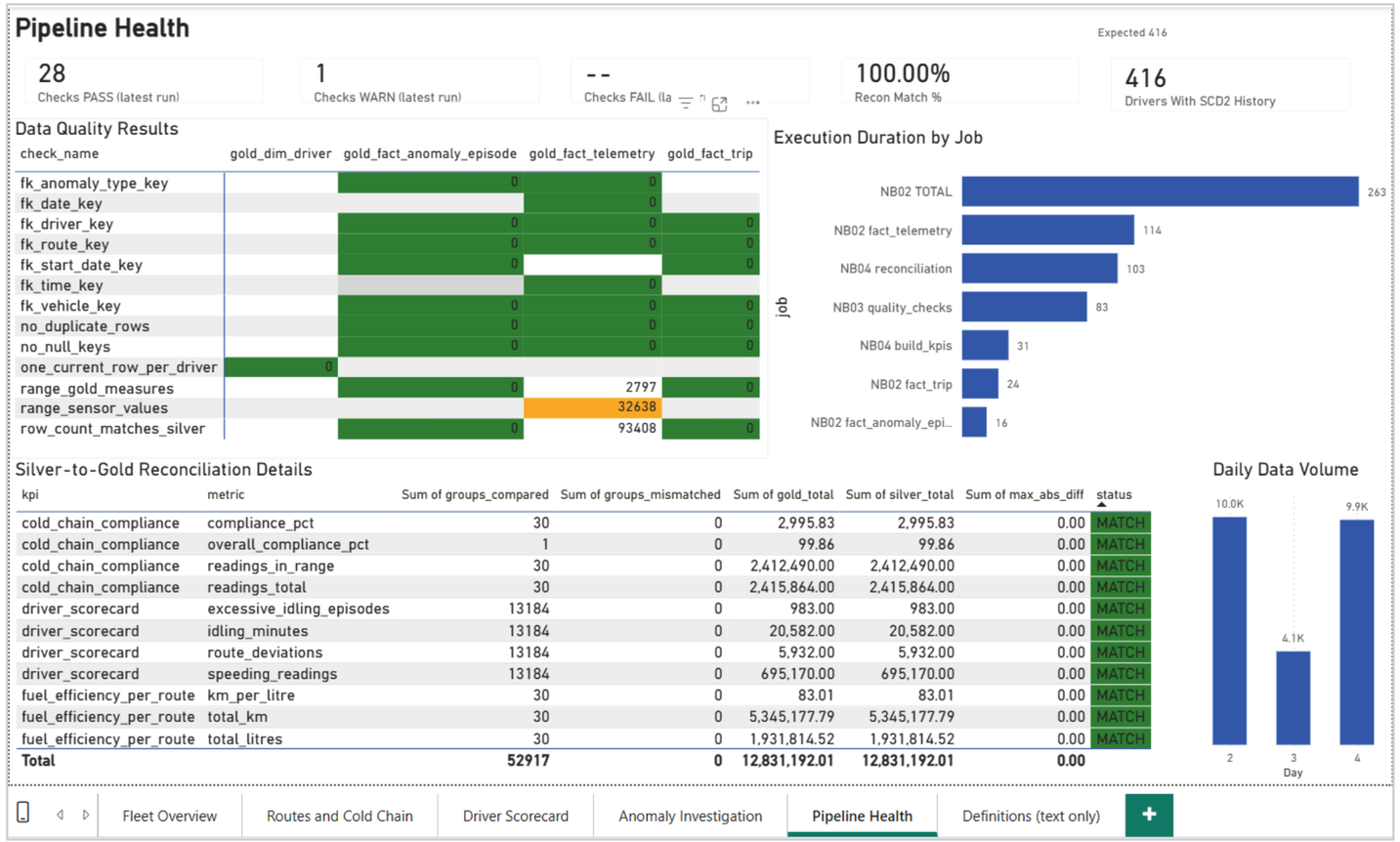


Refresh schedule

Silver anomaly detection → **NB01 Dimensions** → **NB02 Facts** → **NB03 Quality checks** → **NB04 KPIs + reconciliation** → **Power BI refresh**

Note: the two schedules are independent. Check both run histories to confirm the report shows the latest processed data. A blocking quality FAIL stops the pipeline.

Data Quality Gate and Reconciliation



Power BI page 5: Pipeline Health

28 · 1 · 0

checks PASS · WARN · FAIL

100.00%

Silver-to-Gold match, 52,917 groups

416 / 416

drivers with SCD Type 2 history

- Checks: keys, nulls, duplicates, ranges, row counts vs Silver
- NB04 recomputes every KPI from Silver and compares
- The one WARN is by design: out-of-range sensor values stay visible for audit

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Business Metrics at a Glance

2.77 km/L

Fuel efficiency

Fleet-wide distance per litre: 5.35M km on 1.93M L

99.86%

Cold-chain compliance

Refrigerated readings kept within 2–8 °C

163

Cold-chain breach episodes

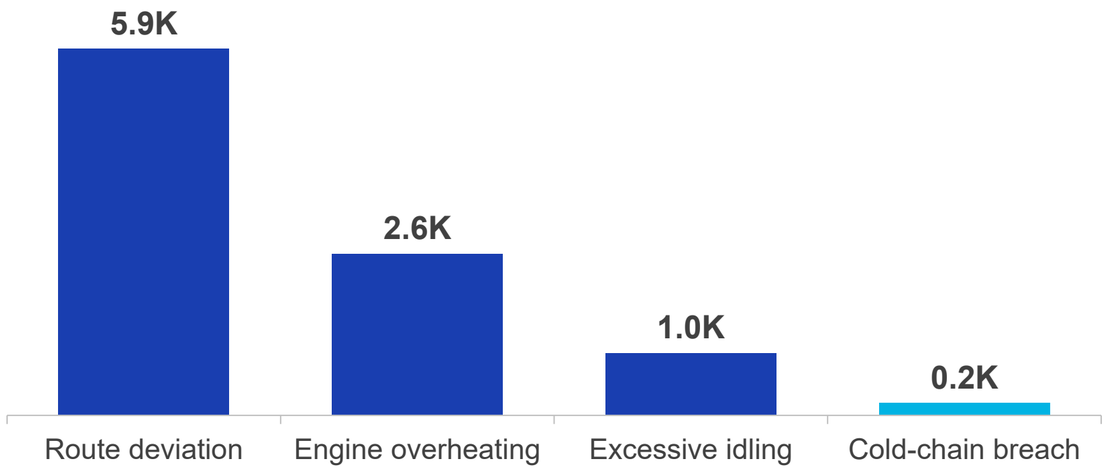
Sustained out-of-range events to investigate

9,724

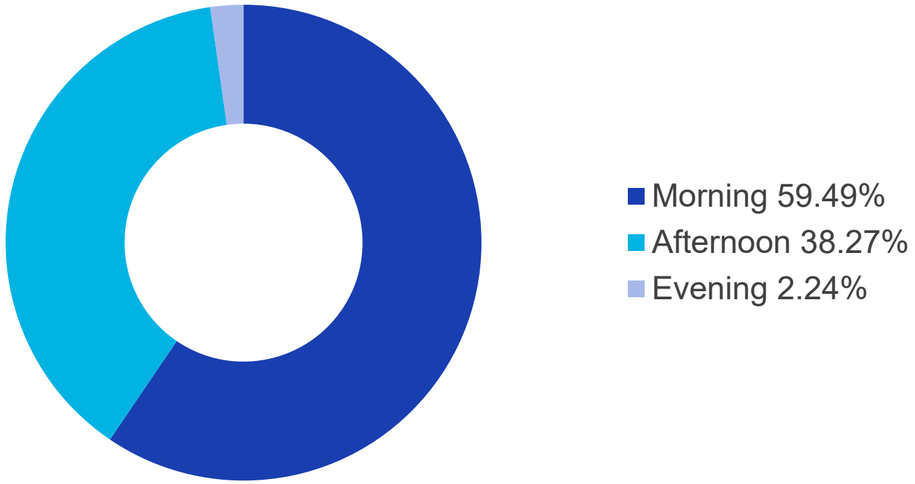
Anomaly episodes

Detected events, each counted once

Anomaly episodes by type (thousands)

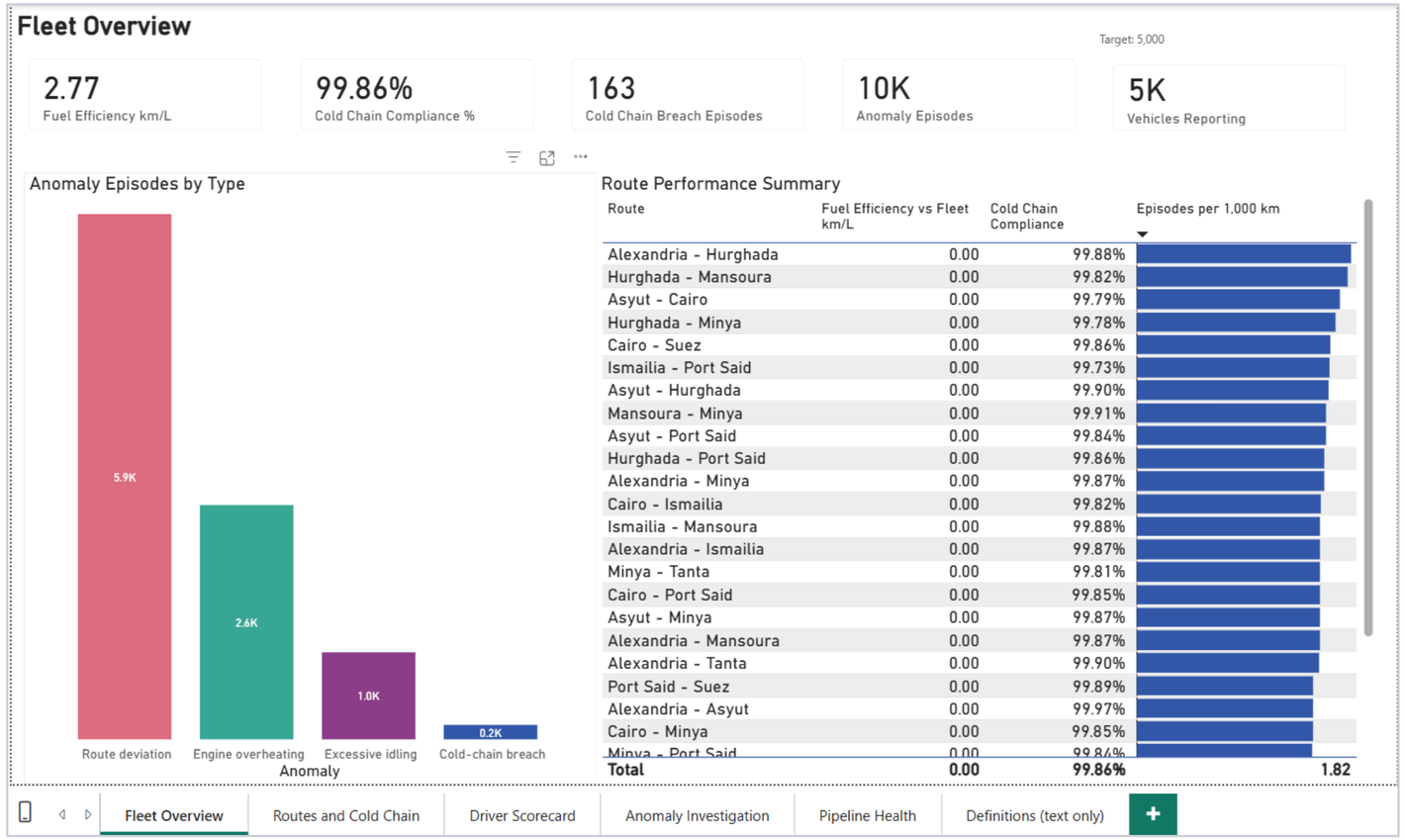


Episodes by time of day



Counts are episodes found by our detection rules on simulated data, not verified real incidents. Values are the documented report snapshot.

Power BI Report: Fleet Overview



Power BI page 1: Fleet Overview

Business meaning: one screen shows fuel use, cold-chain safety and where anomalies cluster.

5,000 vehicles reporting

All simulated vehicles (target 5,000) appear in the Gold layer.

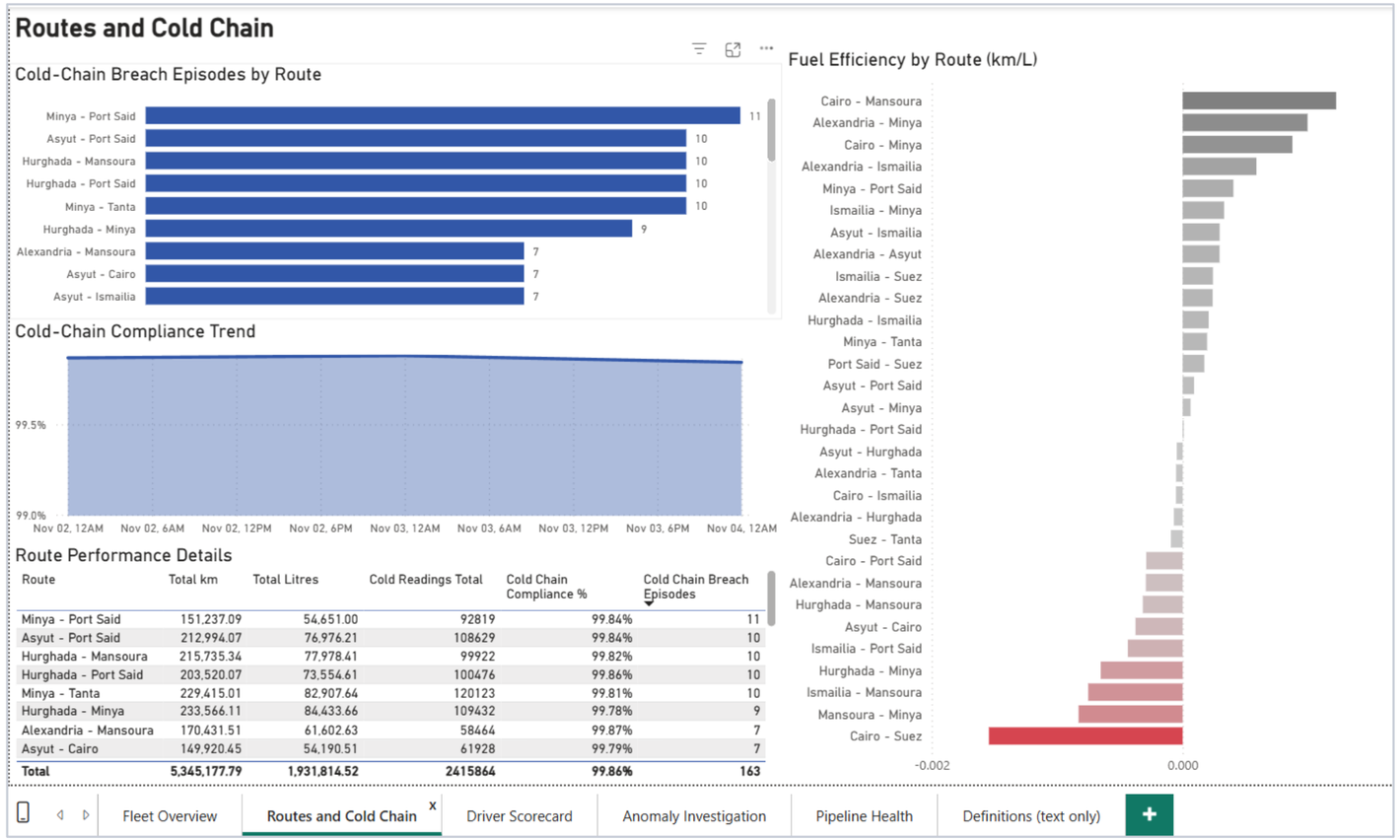
Route performance table

Compare fuel efficiency, cold-chain compliance and anomaly rate per 1,000 km.

Read with care

The “vs Fleet” column shows 0.00 because route gaps are only about ±0.002 km/L.

Power BI Report: Routes and Cold Chain



Power BI page 2: Routes and Cold Chain

Business meaning: managers can target the few routes with repeated breaches, not the whole fleet.

Cold-chain hot spots

Minya–Port Said has the most breach episodes (11). Several routes follow with 9–10.

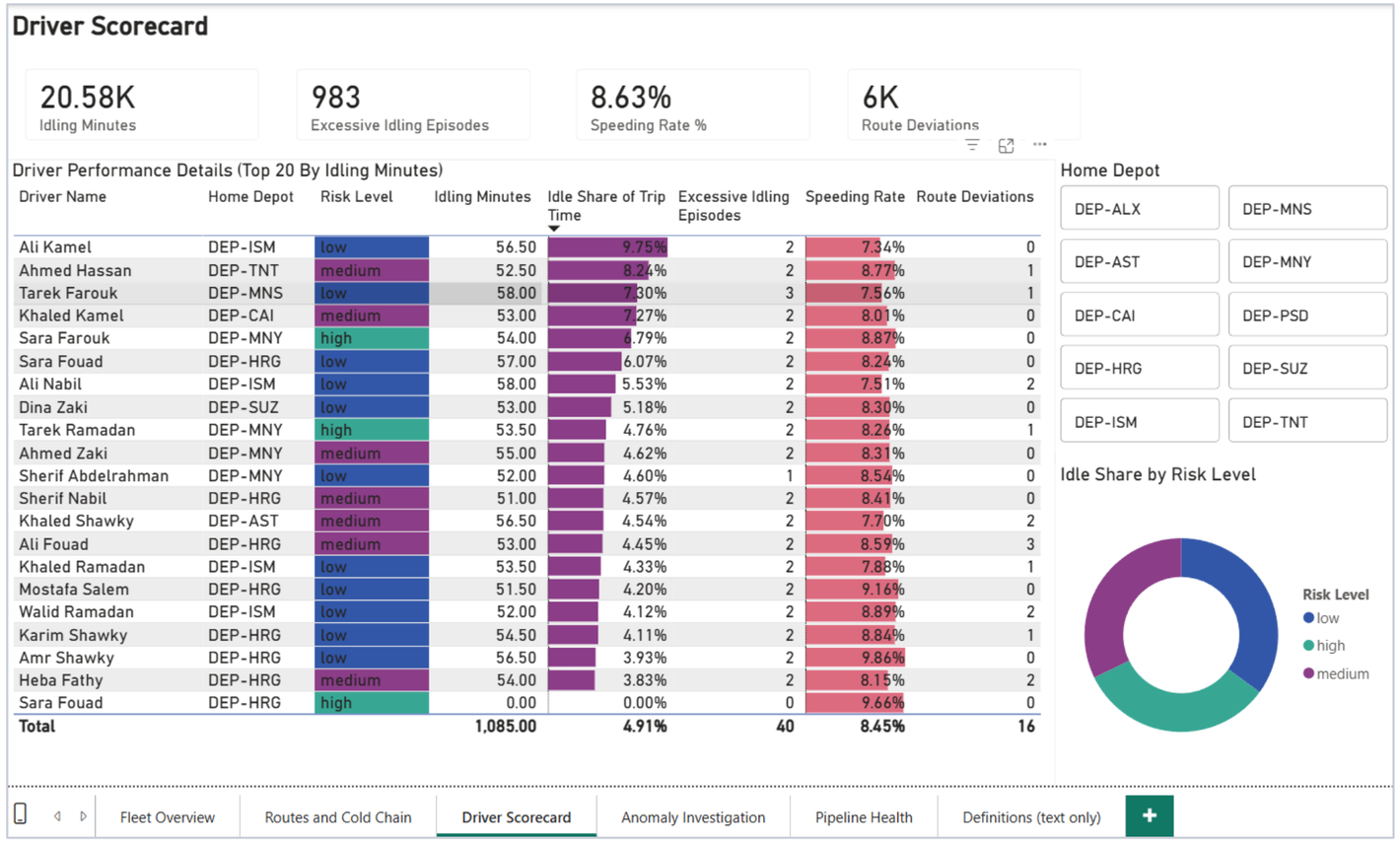
Totals behind 99.86%

2,415,864 refrigerated readings and 163 breach episodes across 30 routes.

Fuel use

5,345,177.79 km on 1,931,814.52 L gives 2.77 km/L. Route gaps are very small.

Power BI Report: Driver Scorecard



Power BI page 3: Driver Scorecard

Business meaning: supervisors can focus coaching on the drivers and depots with the most idling.

Idling

20.58K idling minutes and 983 excessive-idling episodes (10 minutes or more).

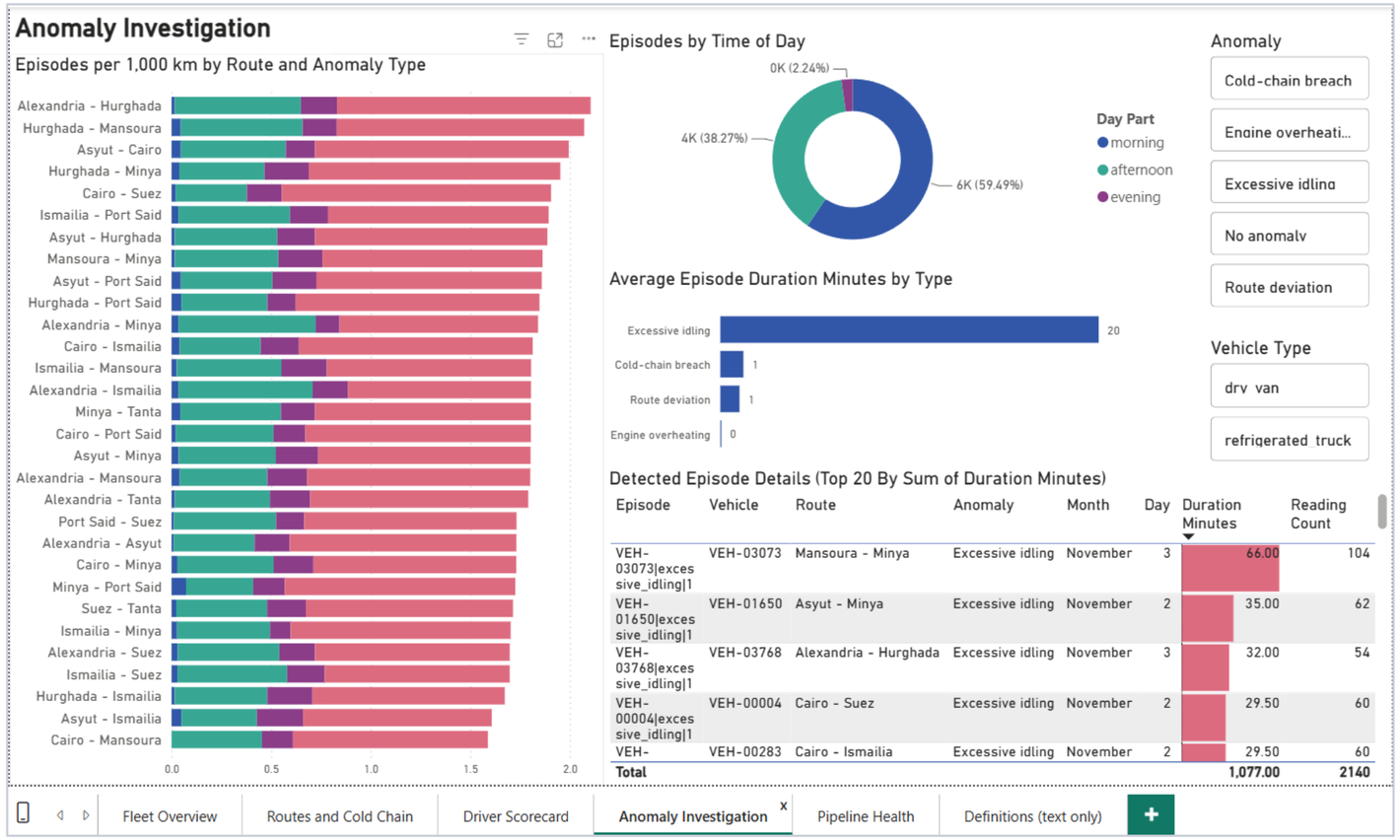
Speeding and routes

8.63% of readings above 90 km/h and 5,932 route-deviation flags.

Drill down

A depot slicer and a Top-20 table show who to coach first. SCD Type 2 keeps depot and risk changes correct.

Power BI Report: Anomaly Investigation



Power BI page 4: Anomaly Investigation

Business meaning: analysts move from “how many” to “where, when and how long”.

How long

Excessive idling lasts about 20 minutes on average. Other types last about 1 minute.

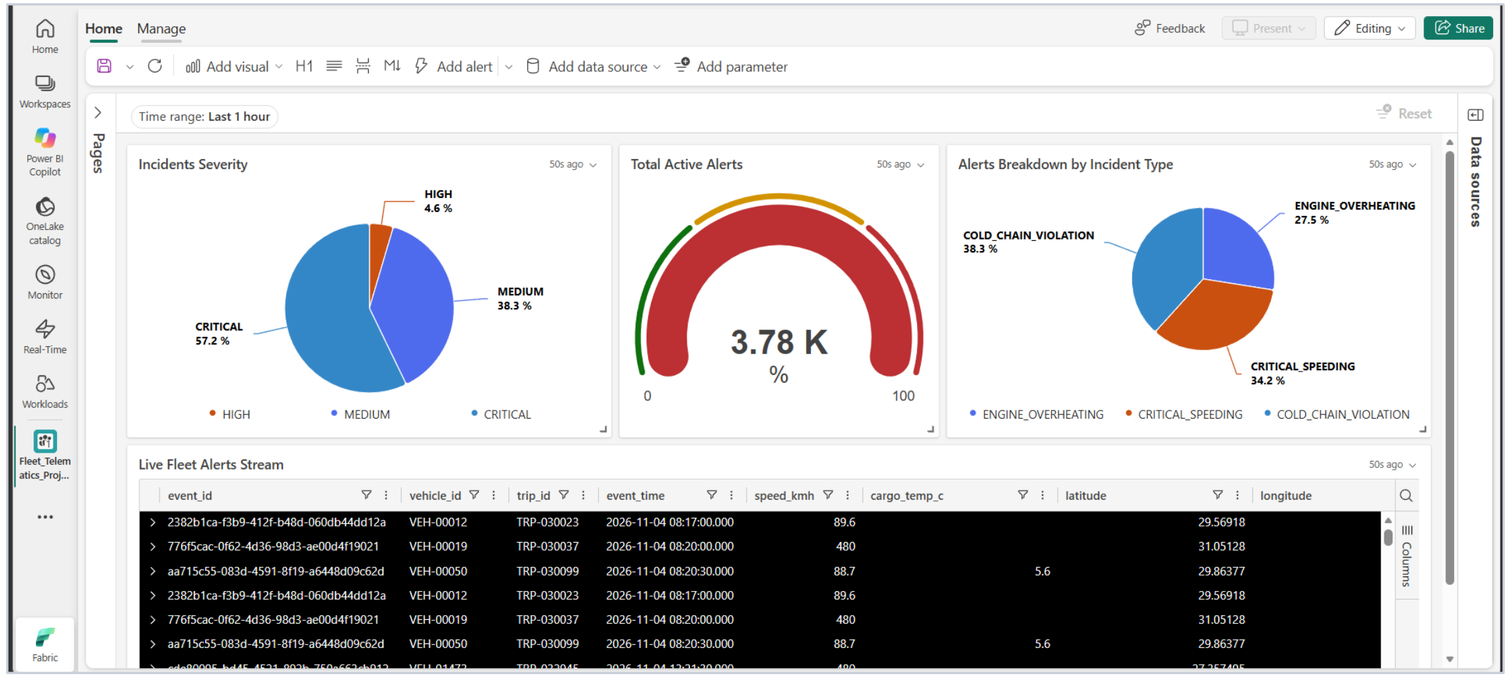
Where

Stacked bars show anomaly intensity per route and type, per 1,000 km.

Longest episode

66 minutes of excessive idling on VEH-03073 (Mansoura–Minya).

Fabric Real-Time Fleet Alerts Dashboard



Microsoft Fabric Real-Time dashboard (time range: last 1 hour)

Incident severity

Critical 57.2%
Medium 38.3%
High 4.6%

Alerts by incident type

Cold-chain violation 38.3%
Critical speeding 34.2%
Engine overheating 27.5%

Power BI report

Gold data, refreshed daily. For trends, routes, drivers and pipeline trust.

Real-time dashboard

Alerts as events arrive, with vehicle, trip, time, speed, cargo temperature and GPS.

Validation: Measured Evidence

Streaming golden run (day 2026-11-04)

3,157,802
events in 75.2 min at 700 events/s

1.59 s
average database-to-cloud latency (p95 2.36 s)

0 lost · 0 duplicated
across a deliberate bridge restart

9,353
late events injected and handled

Batch ingestion: five required runs

Scenario	Inserted	Skipped
1 Initial load	4,014,197	0
2 Incremental #1	293,897	0
3 Incremental #2 + re-run	279,150	293,897
4 Failure and retry	502,818	0
5 Backfill (loaded hour)	0	424,918

Skipped rows are duplicates caught by db_id: re-runs are safe.

Row counts match at every hop (streaming run)



Batch total: 5,090,062 rows loaded; 69,971 bad rows rejected and kept with reasons. Stream: 43,340 rejected at the API. All figures are from simulated data on shared Fabric trial capacity.

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Conclusion: What the Platform Delivers

Fuel efficiency

2.77 km/L

Correct fleet-wide ratio for any filter

Cold chain

99.86%

163 breach episodes to investigate

Driver behaviour

983

excessive-idling episodes; 8.63% speeding

Anomalies

9,724

episodes, each counted once

Pipeline trust

100%

Gold matches Silver; 28 PASS, 1 WARN, 0 FAIL

From raw telemetry to trusted answers

Streaming and batch ingestion, a Medallion lakehouse, quality gates and two dashboards, with measured evidence at every step.

Q&A

Thank you

Questions are welcome

Enterprise Fleet Telematics Lakehouse & Dynamic Logistics Warehouse

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